❖ Git & GitHub

- Introduction to Git & GitHub

1. What is Version Control System (VCS)
   - A **Version Control System (VCS)** is a software tool that tracks changes made to files over time. It allows developers to manage different versions of their code, collaborate with others, and restore previous versions whenever needed.

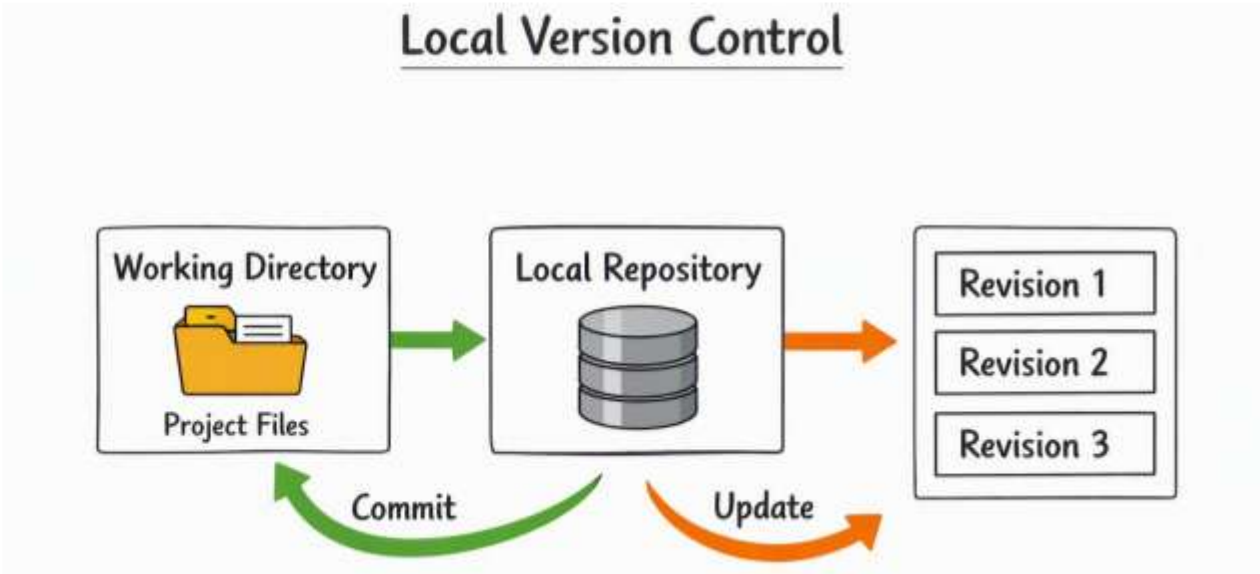
   - A Version Control System keeps the complete history of your project.

2. Why Version Control?
   - Without Version Control
     - Cannot track changes
     - No backup
     - Team collaboration becomes difficult
     - Cannot restore old code
     - Code gets overwritten
     - Difficult to know who changed what

3. Types of Version Control System
   1) Local Version Control System
   2) Centralized Version Control System
   3) Distributed Version Control System (Git)

1) Local Version Control System.
   - A Local VCS stores all versions only on your own computer.

```
+----------------------+
|      Developer       |
|                      |
|  Project Versions    |
|  v1                  |
|  v2                  |
```



Advantages

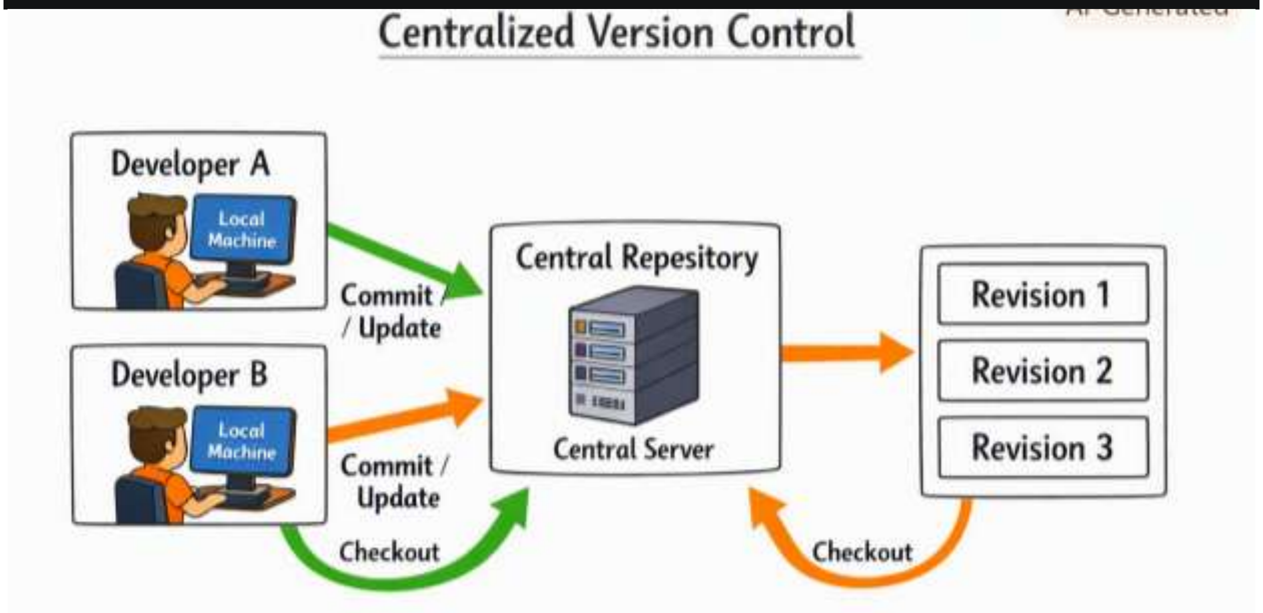
- Easy to use
- Fast
- No internet required

Disadvantages

- Computer crashes → data lost
- No collaboration
- No backup
- Only one developer can work

2) Centralized Version Control System
   - Everyone connects to one server.
   - **Central Repository (Server)** where all code versions are stored

```
             Central Server
          +------------------+
          |    Repository    |
          +------------------+
           /       |        \
          /        |         \
         /         |          \
Developer A  Developer B  Developer C
```



Advantages

- Team collaboration
- Central backup
- Easy management

Disadvantages

If server crashes

↓

Entire project stops.

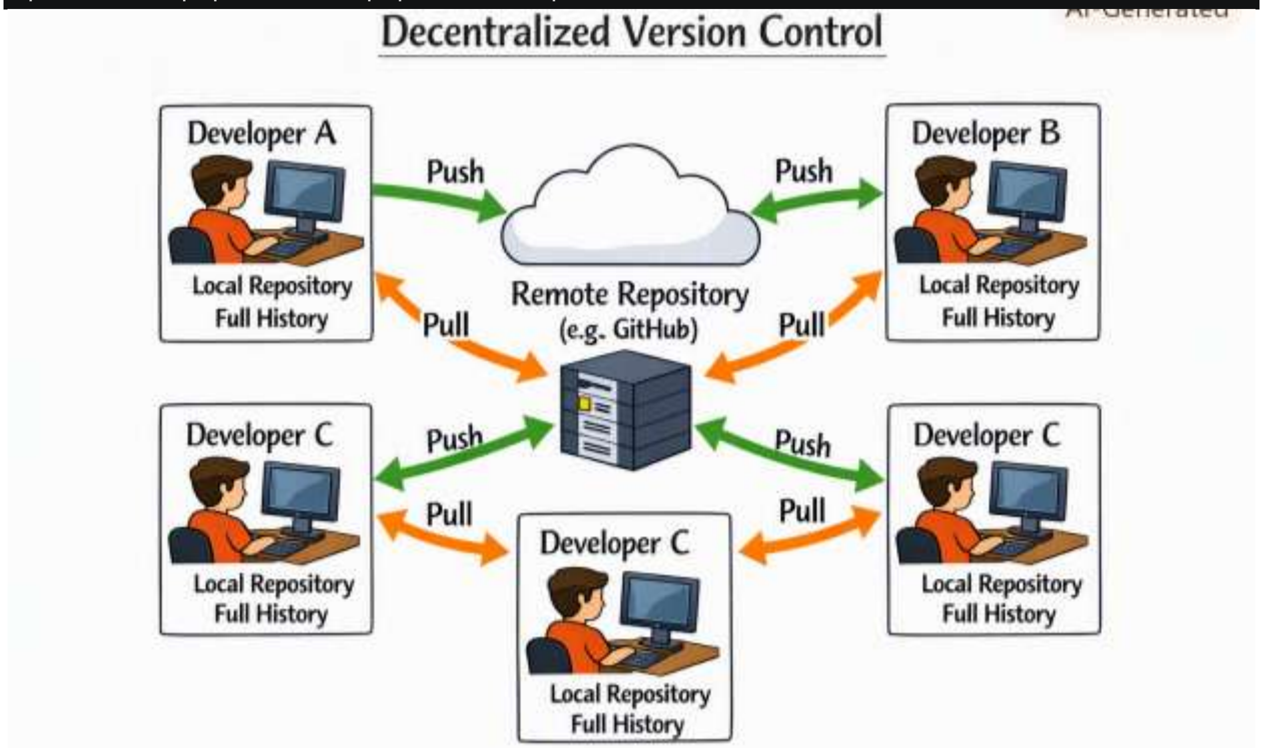
No one can work.

Single Point of Failure.

3) Distributed Version Control System (Git)

- Every developer has a complete copy of the project.
- collaboration is flexible.

```
             GitHub
        +---------------+
        | Remote Repo   |
        +---------------+
          ↑    ↑    ↑
      Push|    |Pull|
+---------+ +---------+ +---------+
| Dev A   | | Dev B   | | Dev C   |
| Local   | | Local   | | Local   |
| Repo    | | Repo    | | Repo    |
+---------+ +---------+ +---------+
```



Advantages

- Full backup
- Offline work
- Fast
- Safe
- Easy collaboration

4. What is Git?
   - Git is a Distributed Version Control System that tracks changes in files and helps developers collaborate efficiently.

   - Git remembers every change made in your project.

   - Git is the most popular Version Control System in the world.

Features of Git

- Fast
- Distributed
- Secure
- Branching
- Merging
- Offline Support
- Version Tracking
- Collaboration
- Easy Rollback

5. What is GitHub?
   - GitHub is a cloud platform that stores Git repositories online.
   - GitHub is like Google Drive for Git projects.

Git	**GitHub**
Software	Cloud Platform
Installed locally	Runs in browser
Tracks versions	Stores repositories
Works offline	Requires internet for sync
Open source	Provides collaboration tools

6. Working Without Git vs With Git

   1) Without Git

Problems:

- Multiple copies
- Lost changes
- Difficult collaboration
- No history
- Manual backups

   2) With Git & GitHub

   Advantages:

- Single source of truth
- Full history
- Easy collaboration
- Safe backups
- Code review with pull requests

7. Git Installation
   - https://git-scm.com/install/windows
   - Install with default settings.
   - git –version

8. GitHub Account
   - Go to **https://github.com**
   - Click **Sign up**
   - Complete your profile.

9. Git Configuration
   - git config --global user.name "Name"
   - git config --global user.email " example @gmail.com"
   - git config --global –list

10 . First Git Workflow

- Step 1
    - Create a folder.
    - MyProject

- Step 2
    - Initialize Git.
    - git init

- Step 3
    - Check status.
    - git status

- Step 4
    - Create a file.
    - Pr.py

- Step 5
    - Stage the file.
    - git add .

- Step 6
    - Commit the changes.
    - git commit -m "Initial commit"

- Step 7
    - Create a new repository on GitHub.

- Step 8
    - Connect the local repository to GitHub.
    - git clone https://github.com/username/MyProject.git

- Step 9
    - Push the code.
    - git push -u origin main